



Laxatives and Anti-diarrhoeal Agents — MCQ Practice

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which laxative should be avoided in pregnancy because it can stimulate uterine contractions?

EASY

- A. Docusate sodium
- B. Lactulose
- C. Castor oil
- D. Psyllium
- E. Polyethylene glycol

2. What is the mechanism of bulk laxatives like psyllium?

EASY

- A. They irritate colonic nerve fibers directly
- B. They form hydrophilic gels that retain water and cause intestinal distension, increasing peristalsis
- C. They inhibit chloride channels
- D. They osmotically draw water via unabsorbed disaccharides
- E. They act as surfactants to soften stool

3. Diphenoxylate and loperamide are structurally related to which class of drugs?

EASY

- A. Benzodiazepines
- B. Meperidine (opioid) analogues
- C. Beta-blockers
- D. 5-HT₃ antagonists
- E. Muscarinic antagonists

4. Which laxative class is not associated with tolerance or dependency, making it suitable for chronic constipation?

EASY

- A. Stimulant laxatives (e.g. senna)
- B. Chloride channel activators (e.g. lubiprostone)
- C. Saline laxatives
- D. Irritant laxatives
- E. All laxative classes carry equal dependency risk

5. An immobile elderly patient is started on psyllium (a bulk laxative) for constipation and develops worsening abdominal distension and pain. What is the most likely explanation?

MEDIUM

- A. Psyllium caused a disulfiram-like reaction
- B. Bulk laxatives require adequate mobility and fluid intake; in immobile patients, the distension effect can precipitate intestinal obstruction rather than relief
- C. Psyllium is not a real laxative and has no mechanism
- D. This is an expected, harmless side effect requiring no action
- E. Psyllium caused a paradoxical decrease in colonic water content

6. A patient with severe active ulcerative colitis develops profuse diarrhea and is prescribed loperamide without further evaluation. What serious complication is this patient now at risk for?

MEDIUM

- A. Disulfiram-like reaction
- B. Toxic megacolon
- C. Aspiration pneumonia
- D. Hypophosphatemia
- E. QT prolongation

7. A patient takes mineral oil laxative while lying flat in bed before sleep and later develops cough, fever, and dyspnea. What is the most likely mechanism?

MEDIUM

- A. Allergic reaction to mineral oil
- B. Aspiration of mineral oil (which should be taken upright) leading to lipid pneumonia
- C. Electrolyte imbalance from mineral oil
- D. Toxic megacolon from lubricant laxative use
- E. Disulfiram-like reaction

8. A patient with chronic kidney disease is being considered for a magnesium-containing laxative. Why might this be a particular concern in this population?

MEDIUM

- A. Magnesium laxatives cause hepatotoxicity in renal failure
- B. Excess magnesium absorption/retention in renal impairment could lead to hyporeflexia, hypotension, or cardiac arrest from magnesium toxicity
- C. Magnesium laxatives are ineffective in renal failure
- D. Renal failure has no bearing on magnesium-based laxatives
- E. Magnesium laxatives cause disulfiram-like reactions in renal failure

9. A patient is prescribed lactulose for constipation. Which statement correctly describes its mechanism?

MEDIUM

- A. It is hydrolyzed by gastric acid to form an osmotic laxative
- B. It is a semisynthetic disaccharide that resists GI enzymes but is degraded by colonic bacteria into acids that raise osmotic pressure, causing fluid accumulation and softer stool
- C. It is a stimulant laxative acting directly on colonic nerve fibers
- D. It is a bulk-forming agent requiring high fluid intake
- E. It is a surfactant that emulsifies with stool

10. A patient with undiagnosed abdominal pain requests a laxative for presumed constipation before any workup. What is the concern with prescribing a laxative in this scenario?

MEDIUM

- A. Laxatives are always safe regardless of undiagnosed abdominal pain
- B. Laxatives are contraindicated with undiagnosed abdominal pain or possible IBD/obstruction due to perforation risk
- C. This concern only applies to stimulant laxatives, not other classes
- D. Undiagnosed abdominal pain is not a contraindication for any GI drug
- E. The only risk is a mild allergic reaction

11. A patient taking a poorly-absorbed, extended-release oral medication is started on a stimulant laxative for constipation. What pharmacokinetic interaction should be anticipated?

MEDIUM

- A. No interaction, since laxatives only affect the colon
- B. The laxative may reduce the effect of the extended-release drug by speeding it through the gut before full absorption occurs
- C. The laxative would increase absorption of the extended-release drug
- D. This combination would cause hepatotoxicity
- E. This combination is specifically dangerous only with bulk laxatives

12. A patient with traveler's diarrhea is prescribed bismuth subsalicylate and later reports black stools and a black-appearing tongue. How should this be interpreted?

MEDIUM

- A. This indicates GI bleeding requiring urgent endoscopy
- B. This is an expected, benign adverse effect of bismuth subsalicylate, not a sign of bleeding
- C. This indicates disulfiram-like reaction
- D. This indicates toxic megacolon
- E. This is a sign of aspiration

13. A physician wants to select a laxative for a patient needing bowel preparation before colonoscopy, requiring rapid, complete colonic evacuation. Which laxative type is specifically noted for this purpose?

MEDIUM

- A. Docusate sodium (stool softener)
- B. Polyethylene glycol (PEG) electrolyte solution, used as colonic lavage before radiologic/endoscopic procedures
- C. Mineral oil
- D. Bulk laxatives like psyllium
- E. Bisacodyl suppository alone

14. A patient started on docusate sodium for prophylactic stool softening after surgery is frustrated that it hasn't worked after one dose the night before their bowel movement was due. What should they be told?

MEDIUM

- A. Docusate sodium works within minutes and this is a treatment failure requiring a different drug
- B. Stool softeners like docusate take days to work and are meant for prophylaxis, not acute/rapid treatment
- C. Docusate sodium is not a real laxative
- D. This indicates an allergic reaction requiring drug discontinuation
- E. Docusate should always be combined with mineral oil for any effect

15. A patient with chronic constipation has failed adequate trials of fiber, fluids, and exercise, and is now being escalated through pharmacologic options. The physician wants to avoid both the dependency risk of stimulant laxatives and the obstruction risk of bulk laxatives (the patient has limited mobility). Which laxative class best fits this constraint?

HARD

- A. Stimulant laxatives, since they work fastest
- B. Chloride channel activators (lubiprostone), since they are not associated with tolerance/dependency and don't rely on bulk-forming distension that risks obstruction in immobile patients
- C. Bulk laxatives at a higher dose
- D. Emollient laxatives alone, since they have no risk profile at all
- E. Saline laxatives used chronically

16. A researcher compares senna (stimulant) and lactulose (osmotic) mechanistically. Senna causes water/electrolyte secretion into the bowel directly via stimulant action, while lactulose's osmotic effect depends on bacterial metabolism. In a patient who has recently completed a course of broad-spectrum antibiotics that suppressed colonic flora, which laxative's efficacy would be more directly compromised, and why?

HARD

- A. Senna, because stimulant laxatives require intact bacterial flora
- B. Lactulose, because its osmotic effect specifically depends on colonic bacteria degrading it into osmotically active acids — suppressed flora would blunt this metabolic step
- C. Neither would be affected, since antibiotics have no bearing on laxative mechanisms
- D. Both would be equally and severely compromised
- E. Bisacodyl would be the only laxative affected

17. A patient on chronic opioid therapy for cancer pain develops opioid-induced constipation. Considering that diphenoxylate/loperamide work via presynaptic opioid receptors in the enteric nervous system to REDUCE peristalsis, would adding loperamide help this patient's constipation?

HARD

- A. Yes, loperamide would help since it is technically an "antidiarrheal that affects motility"
- B. No — loperamide works by the same general opioid-receptor mechanism that is already causing the opioid-induced constipation, so adding it would be pharmacologically counterproductive and could worsen constipation
- C. Loperamide has no relevant mechanism for this scenario at all
- D. Loperamide would reverse opioid-induced constipation via a competitive antagonist mechanism
- E. This combination is required for adequate pain control

18. A clinician considers combining bisacodyl (stimulant, acts directly on colonic nerve fibers) with docusate sodium (surfactant/stool softener) for a postoperative patient. What is the pharmacological rationale for this combination rather than either alone?

HARD

- A. The combination is pharmacologically redundant and pointless, since both work identically
- B. The two act by complementary mechanisms — bisacodyl stimulates peristalsis directly while docusate softens the stool itself, addressing both motility and stool consistency, a rationale consistent with common postoperative bowel regimens to avoid straining
- C. Docusate would completely block bisacodyl's mechanism, making the combination contraindicated
- D. This combination would cause a disulfiram-like reaction
- E. Only the surfactant component has any effect; the stimulant component does nothing when combined

19. A patient with chronic laxative overuse (stimulant laxatives for years) presents with muscle weakness and is found to have significant electrolyte derangement. Which underlying principle from the lecture explains this presentation, and what does it imply about long-term laxative use generally?

HARD

- A. Laxatives have no systemic effects regardless of duration of use
- B. Chronic laxative use risks electrolyte imbalance (as explicitly noted for laxative use generally) plus dependency, which is why laxatives are indicated only for short-term constipation management, not chronic first-line therapy
- C. This presentation is unrelated to laxative use and must be coincidental
- D. Only bulk laxatives cause electrolyte imbalance; stimulants are exempt
- E. Electrolyte imbalance only occurs with saline laxatives, never with stimulants

20. A patient with IBS-related chronic constipation who also has a documented allergy to sulfa-containing compounds is being considered for multiple laxative classes. A colleague raises concern about cross-reactivity with a chloride channel activator. Reviewing the lecture's description of lubiprostone's metabolism (rapid metabolism in the stomach and jejunum, minimal drug-drug interactions), what is the most defensible clinical reasoning?

HARD

- A. Lubiprostone should be avoided in all allergy patients regardless of allergen, since caution is never warranted
- B. Lubiprostone's described profile (rapid local GI metabolism, minimal systemic drug-drug interaction) does not inherently establish sulfa cross-reactivity; this specific allergy concern would need direct verification against lubiprostone's actual chemical structure rather than assumed from its pharmacokinetic profile alone
- C. All laxatives are chemically related to sulfonamides and must be avoided in sulfa allergy
- D. Sulfa allergy is only relevant to antibiotics and never to any GI drug, so no further thought is needed
- E. Lubiprostone's minimal interaction profile means it is definitely safe in any allergy without exception

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	Castor oil is broken down to ricinoleic acid, which can stimulate uterine contractions.
2	B	EASY	Bulk laxatives form hydrophilic gels that retain water, causing distension and increased peristalsis.
3	B	EASY	Both are meperidine analogues with opioid-like action on the gut.
4	B	EASY	Lubiprostone (chloride channel activator) is specifically noted as not being linked to tolerance or dependency.
5	B	MEDIUM	Bulk laxatives should be used cautiously in immobile patients due to the risk of intestinal obstruction from the distension effect without adequate motility.
6	B	MEDIUM	

Antimotility agents like loperamide are contraindicated in severe colitis due to the risk of precipitating toxic megacolon.

7	B	MEDIUM	Mineral oil must be taken upright to avoid aspiration and the resulting risk of lipid pneumonia.
8	B	MEDIUM	Excess magnesium (as can occur with impaired renal clearance) risks hyporeflexia, hypotension, and cardiac arrest.
9	B	MEDIUM	Lactulose is not hydrolyzed by GI enzymes but is degraded by colonic bacteria into lactic, formic, and acetic acids, raising osmotic pressure and drawing fluid into the colon.
10	B	MEDIUM	Laxatives are contraindicated in undiagnosed abdominal pain or IBD due to the risk of intestinal perforation, and in organic GIT obstruction.
11	B	MEDIUM	Laxatives accelerate GI transit, which can reduce the effect of poorly-absorbed, delayed-acting, or extended-release oral drugs by speeding them through before absorption completes.
12	B	MEDIUM	Black tongue and black stools are the noted characteristic (benign) adverse effects of bismuth subsalicylate.
13	B	MEDIUM	PEG electrolyte solutions are specifically used for colonic lavage before radiologic/endoscopic procedures.
14	B	MEDIUM	Stool softeners take days to work and are used for prophylaxis, not acute treatment.
15	B	HARD	Lubiprostone avoids both concerns: it is not associated with tolerance/dependency (unlike stimulants) and works via chloride channel activation rather than bulk-forming distension (avoiding the immobile-patient obstruction risk of bulk laxatives).
16	B	HARD	Lactulose requires colonic bacteria to degrade it into lactic/formic/acetic acids for its osmotic effect; suppressed flora from recent antibiotics would blunt this specific mechanism, while senna's direct stimulant/secretory action on the bowel wall does not depend on bacterial metabolism.
17	B	HARD	Loperamide activates the same class of presynaptic opioid receptors in the enteric nervous system that cause opioid-induced constipation from systemic opioids; adding it to a constipated opioid-therapy patient would be mechanistically counterproductive rather than helpful.
18	B	HARD	Bisacodyl (direct colonic stimulant) and docusate (stool softener/surfactant) act via distinct, complementary mechanisms — stimulating motility versus softening stool consistency — supporting their combined use, particularly postoperatively to ease passage and avoid straining.
19	B	HARD	Chronic laxative use (generally, and stimulants specifically carry dependency risk) risks electrolyte imbalance — this is why laxatives are indicated for short-term use, with non-drug measures as first-line and prolonged use to be avoided.
20	B	HARD	The lecture describes lubiprostone's pharmacokinetic profile (rapid gastric/jejunal metabolism, minimal interactions) but this does not by itself address allergen cross-reactivity, which depends on chemical structure rather than metabolic profile — illustrating the importance of not conflating pharmacokinetic favorability with allergy safety.